

BSc CSIT 2nd Semester Board Exam Study Plan

July 16 (Asar 30) - August 15 (Shrawan 30) | Balanced 30-Day Preparation Plan

Day 1 - July 16 (Nepali: Asar 30)

Statistics I:

- Intro + Types of data + Population/sample

Discrete Structures:

- Sets + Subsets + Venn Diagram

Mathematics II:

- Linear Equations basic + Matrix form

Day 2 - July 17 (Nepali: Shrawan 1)

Statistics I:

- Central Tendency - Mean, Median

Discrete Structures:

- Set operations, Inclusion-Exclusion

Mathematics II:

- Row reduction, Echelon forms

Day 3 - July 18 (Nepali: Shrawan 2)

Statistics I:

- Dispersion - Range, SD, CV

Discrete Structures:

- Functions - Injective/Surjective

Microprocessor:

- What is Microprocessor, Bus Organization

Day 4 - July 19 (Nepali: Shrawan 3)

Statistics I:

- Skewness, Kurtosis, Moments

Discrete Structures:

- Inverse, Composite Functions, Floor/Ceiling

Microprocessor:

- ALU, Registers, Control & Timing

Day 5 - July 20 (Nepali: Shrawan 4)

Statistics I:

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- Probability basics, Laws of probability

Discrete Structures:

- Sequences & Summations

Microprocessor:

- 8085 Architecture basics

Day 6 - July 21 (Nepali: Shrawan 5)

Statistics I:

- Bayes Theorem, Simple Qs

Mathematics II:

- Vector Equations, Matrix Eq. $Ax = b$

OOP:

- Structured vs OOP, Features of OOP

Day 7 - July 22 (Nepali: Shrawan 6)

Statistics I:

- Sampling, Errors, Types

Mathematics II:

- Linear Independence, Applications

OOP:

- C++ Program structure, Tokens, Data Types

Day 8 - July 23 (Nepali: Shrawan 7)

Statistics I:

- Random Variables

Discrete Structures:

- Integers, Euclidean Algorithm

Mathematics II:

- Matrix Operations

Day 9 - July 24 (Nepali: Shrawan 8)

Statistics I:

- Expectation, Simple Qs

Discrete Structures:

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- Number Theory Applications

Mathematics II:

- Inverse of matrix, Partitioned matrices

Day 10 - July 25 (Nepali: Shrawan 9)

Statistics I:

- Bernoulli, Binomial Distribution

Discrete Structures:

- Logic - Propositional Logic

Microprocessor:

- Instruction Cycle basics

Day 11 - July 26 (Nepali: Shrawan 10)

Statistics I:

- Poisson & Normal Distribution

Discrete Structures:

- Propositional Equivalence

Microprocessor:

- Fetch/Execute cycle & Timing

Day 12 - July 27 (Nepali: Shrawan 11)

Statistics I:

- Joint Probability, Exponential, Gamma

Discrete Structures:

- Predicates, Quantifiers, Rules of Inference

Microprocessor:

- Memory, I/O Devices basics

Day 13 - July 28 (Nepali: Shrawan 12)

Statistics I:

- Correlation - Karl Pearson, Spearman

Discrete Structures:

- Proof Methods - Direct, Indirect

OOP:

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- Functions, Overloading, Inline

Day 14 - July 29 (Nepali: Shrawan 13)

Statistics I:

- Regression, Least Squares

Discrete Structures:

- Proof by Contradiction/Contraposition

OOP:

- Pointers & Arrays

Day 15 - July 30 (Nepali: Shrawan 14)

Discrete Structures:

- Induction, Strong Induction

Mathematics II:

- Linear Transformations

OOP:

- Classes & Objects basics

Day 16 - July 31 (Nepali: Shrawan 15)

Discrete Structures:

- Recursion, Recursive Algorithms

Mathematics II:

- Matrix of Linear Transformation

OOP:

- Constructors & Destructors

Day 17 - August 1 (Nepali: Shrawan 16)

Discrete Structures:

- Counting Basics, Pigeonhole Principle

Mathematics II:

- Determinants intro, Cramer's Rule

Microprocessor:

- Assembly Language Programming basics

Day 18 - August 2 (Nepali: Shrawan 17)

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Discrete Structures:

- Permutations & Combinations

Mathematics II:

- Properties of Determinants, Volume

Microprocessor:

- Instruction Types, Mnemonics, Operands

Day 19 - August 3 (Nepali: Shrawan 18)

Discrete Structures:

- Binomial Coefficients, Generating Permutations

Mathematics II:

- Vector Spaces & Subspaces

OOP:

- Operator Overloading - Unary & Binary

Day 20 - August 4 (Nepali: Shrawan 19)

Discrete Structures:

- Discrete Probability, Expected Value, Variance

Mathematics II:

- Null Space, Column Space, Basis

OOP:

- Data Conversion (basic user-defined)

Day 21 - August 5 (Nepali: Shrawan 20)

Discrete Structures:

- Recurrence Relations, Divide & Conquer

Mathematics II:

- Dimension, Rank, Change of Basis

Microprocessor:

- Addressing Modes, Simple Programs

Day 22 - August 6 (Nepali: Shrawan 21)

Discrete Structures:

- Relations - Properties, Representation

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Mathematics II:

- Eigenvalues, Eigenvectors intro

Microprocessor:

- Memory Read/Write, I/O Ops

Day 23 - August 7 (Nepali: Shrawan 22)

Discrete Structures:

- Equivalence Relations, Partial Ordering

Mathematics II:

- Characteristic Equations, Diagonalization

Microprocessor:

- DMA & Interrupts

Day 24 - August 8 (Nepali: Shrawan 23)

Discrete Structures:

- Graph Basics, Graph Types & Models

Mathematics II:

- Complex Eigenvalues, Applications

OOP:

- Inheritance - Types, Base & Derived

Day 25 - August 9 (Nepali: Shrawan 24)

Discrete Structures:

- Graph Connectivity, Euler & Hamiltonian

Mathematics II:

- Orthogonality, Inner Product

OOP:

- Constructors in Derived Classes, Aggregation

Day 26 - August 10 (Nepali: Shrawan 25)

Discrete Structures:

- Matching, Shortest Path, Dijkstra

Mathematics II:

- Gram-Schmidt, Least Squares

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OOP:

- Virtual Functions, Abstract Classes

Day 27 - August 11 (Nepali: Shrawan 26)

Discrete Structures:

- Trees - Traversals, Spanning Trees

Microprocessor:

- 8259A, 8255A

OOP:

- Polymorphism, Virtual Base Class

Day 28 - August 12 (Nepali: Shrawan 27)

Discrete Structures:

- Minimum Spanning Tree (Kruskal's Algorithm)

Microprocessor:

- 8086 Block Diagram, Segments

OOP:

- Friend Function, This Pointer

Day 29 - August 13 (Nepali: Shrawan 28)

Discrete Structures:

- Network Flows, Max Flow-Min Cut

Microprocessor:

- 80286, 80386, Protected Mode, Paging

OOP:

- Function Templates, Exception Handling

Day 30 - August 14 (Nepali: Shrawan 29)

Microprocessor:

- Final Assembly Qs & Diagrams

OOP:

- File I/O, Stream Class Hierarchy, Random Access

Day 31 - August 15 (Nepali: Shrawan 30)

Revision & Practice:

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- Old Question Papers
- Weak Topics Review
- Flashcards & Formulae